

Harshit Raja

Senior Software Developer

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Professional Summary

Senior Software Developer with **2.7+ years of experience** in designing, developing, and deploying scalable web, desktop, and cloud-based solutions. Skilled in **.NET, C#, React, Angular, Node.js, and Python**, with hands-on expertise in **Autodesk Forge/ACC, BIM APIs, and 3D visualization (Three.js)**. Proven track record of delivering complex client projects, building **AI-powered applications**, and leading teams through **system design, API integration, and performance optimization**. Passionate about bridging software engineering with **BIM, automation, and AI-driven solutions**.

Technical Skills

- **Backend:** C#, .NET, Node.js, REST APIs, Microservices, Java (TestNG, Spring basics), Python (NumPy, TensorFlow, Flask, FastAPI)
- **Frontend:** React, Angular, Vue.js, JavaScript, TypeScript, HTML5, CSS3, Tailwind, Bootstrap
- **3D & BIM Tech:** Autodesk Forge APIs, Autodesk ACC, BIM 360, Data Exchange GraphQL, Three.js, Unity (URP & Shader Graph), IFC, LandXML
- **Databases:** SQL Server, MySQL, PostgreSQL, MongoDB
- **Testing & Automation:** Selenium (Java, TestNG, Maven, ExtentReports), RestAssured (API Testing), Appium (Android/iOS, Cucumber BDD)
- **Cloud & DevOps:** Docker, Render, Vercel, GitLab CI/CD, Atlassian JIRA/Confluence, Git/SVN
- **Other:** OAuth 2.0, JWT, System Design, Scalability & Performance Optimization, Microservices Architecture

Professional Experience

Software Developer | *nCircle Tech Pvt. Ltd.* | Oct 2023 – Present

- Designed and developed **C# .NET desktop tools and Angular web applications** to support BIM workflows.
- Built **custom Three.js binary format (LandPack + JSON manifest)** reducing 3D model load time by **40%** compared to GLB.
- Developed **Autodesk Forge Viewer integrations** with **Data Exchange GraphQL API**, enabling real-time BIM data filtering, element isolation, and collaborative chat.
- Led **proof-of-concept (POC) projects** for major clients (AECOM, Autodesk) to validate feasibility of advanced features like **clash detection explanations** and **automated parameter population**.
- Mentored junior developers, conducted **code/design reviews**, and optimized project delivery timelines.

Software Engineer | *BusinessNext (CRMNEXT), Acidaes Solutions Pvt. Ltd.* | Nov 2022 – Sep 2023

- Enhanced **CRM platform modules** using React, Angular, C#, and .NET, improving **customer engagement workflows**.
- Built **sales pipeline visualization dashboards** that improved reporting speed by **30%**.
- Developed **advanced analytics and lead tracking features**, enabling data-driven decision making for clients.
- Collaborated with QA teams to resolve defects, reducing bug resolution time by **25%**.

Software Development Intern | *nCircle Tech Pvt. Ltd.* | Aug 2022 – Nov 2022

- Developed **desktop image analyzer in WPF** to detect architectural drawing discrepancies.
- Implemented efficient **image comparison algorithms**, reducing review time by **50%**.
- Delivered early-stage prototypes and assisted in POCs for **BIM model management solutions**.

Key Projects

AI Chat Bot Interaction with Forge Viewer

Node.js, Express, LangChain, Autodesk Data Exchange API, Socket.IO

- Built a **natural language assistant** integrated into Autodesk Forge Viewer.
- Enabled queries like “*Show me all walls on Level 2*” by generating and executing **GraphQL queries**.
- Implemented **real-time chat context tracking** with WebSockets, syncing BIM model element isolation across users.

PlanetSense – Automated LULC & Object Delineation

Python, TensorFlow, Rasterio, Fiona, .tif/.shp/.gdb/.laz

- Developed a **GUI tool for Land Use/Land Cover classification** and **object extraction** from imagery & LiDAR data.
- Designed **modular pipeline** with support for raster & vector exports.
- Improved processing speed by **35%** using optimized TensorFlow models and parallelized workflows.

AIClipCreator

Python, Streamlit, Whisper, FFmpeg

- Created a **desktop app to auto-generate short-form video clips with AI-powered transcripts and captions**.
- Implemented **speech-to-text (Whisper) integration** for accurate multilingual transcription.
- Added **batch video trimming, auto-caption overlays, and export to vertical 9:16 format** for YouTube Shorts.

Custom Three.js LandPack (.Ixpak) + JSON Manifest

C#, Node.js, Three.js, Autodesk Forge APIs

- Designed a **binary file format** optimized for storing and rendering large BIM models.
- Reduced model size vs traditional formats; achieved **faster parsing and rendering** in Forge Viewer.
- Created **JSON manifest** to store metadata (categories, levels, element properties) for efficient querying.

Forge Viewer – Advanced BIM Interactions

Autodesk Forge Viewer, Node.js, JavaScript, Socket.IO

- Implemented **real-time 3D model filtering, annotations, and measurement tools**.
- Enabled **multi-user chat integration with model context awareness**.
- Enhanced **performance & scalability**, ensuring smooth visualization for large AEC models.

Additional Projects

- **Reels Downloader (Android App)**: Flutter app for downloading Instagram Reels without watermarks & preserving audio (1,000+ downloads).
- **Shopping App (Android)**: eCommerce app with real-time Firebase integration & secure authentication.
- **Hexmail (Discord + Gmail Bot)**: Integrated Gmail with Discord for real-time notifications & filtering.
- **ClipCraft (Desktop Video Processor)**: GUI tool for batch video compression using FFmpeg (60% smaller file size).
- **BIM Manager Console App**: .NET tool for managing BIM360 accounts with efficient CRUD operations.
- **CCT – IFC Viewer & Editor**: Automated conversion of IFC models into optimized 3D assets for faster rendering.

Education

Bachelor of Technology (B.Tech) – Computer Science

Dr. A.P.J. Abdul Kalam Technical University - (Score 80%)

High School & Intermediate

Central Board of Secondary Education (CBSE) - (Score : 80%)

Languages

English (Fluent), **German** (Learning A1), **Russian** (Completed A1 now Learning A2)

Additional Highlights

- Strong foundation in **System Design (scalability, replication, caching, failover, load balancing)**
- Experienced in **CI/CD pipelines** with GitLab & Docker for automated testing and deployment
- Mentored junior developers and conducted **knowledge-sharing sessions**
- Actively building **local AI + BIM prototypes** integrating **Ollama, LangChain, and Postgres**
- Ongoing **DSA & System Design prep for FANG-level interviews**